Silicon(IV) Oxide

Exam code: 0620 & 0971

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Structure of silicon(IV) oxide

What is the structure of silicon(IV) oxide?

Extended tier only

- Silicon(IV) oxide (also known as silicon dioxide or silica), SiO_2, is a macromolecular compound which occurs naturally as **sand** and **quartz**
- Each oxygen atom forms covalent bonds with **2** silicon atoms and each silicon atom in turn forms covalent bonds with **4** oxygen atoms
- A tetrahedron is formed with one silicon atom and four oxygen atoms, similar to diamond

The structure of silicon(IV) oxide

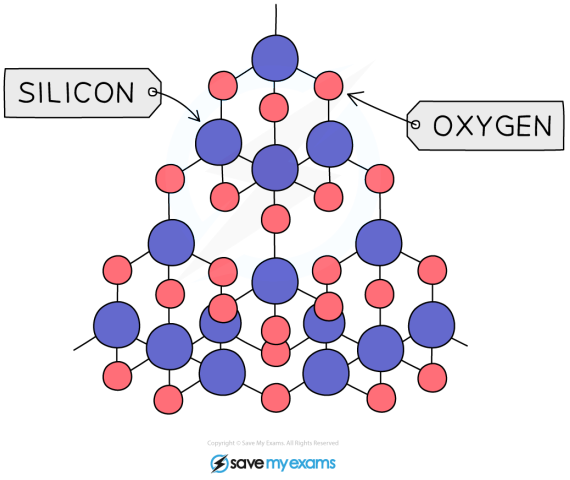

Diagram showing the structure of SiO_2 with the silicon atoms in blue and the oxygen atoms in red

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Ask questions and get help with exactly what you're revising.
Got it

silicon(IV) oxide

How does diamond compare to silicon(IV) oxide?

Extended tier only

- SiO_2 has lots of very strong covalent bonds and no intermolecular forces so it has similar properties to diamond
- It is very **hard**, has a very **high** boiling point, is insoluble in water and does not conduct electricity
- SiO_2 is cheap since it is available naturally and is used to make sandpaper and to line the inside of furnaces

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